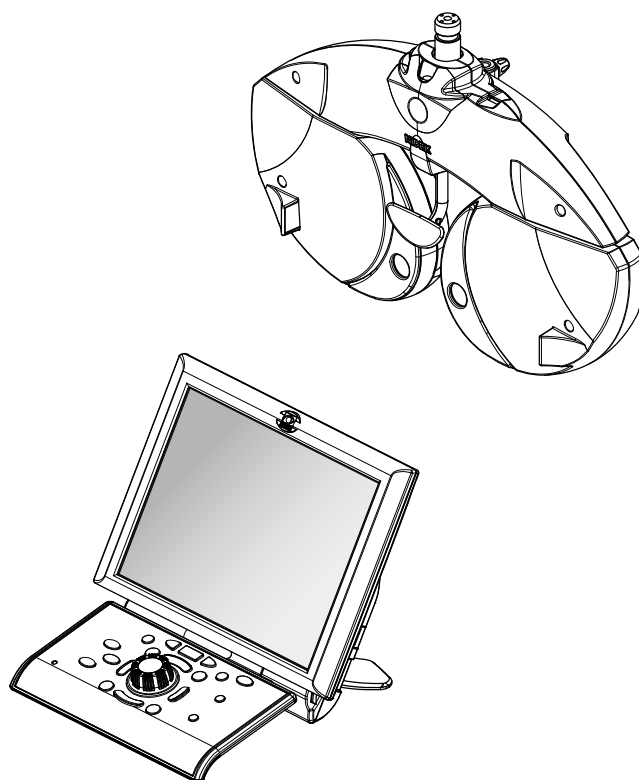


NIDEK

REFRACTOR
RT-6100

INTERFACE MANUAL



Original instructions

NIDEK CO., LTD.

NIDEK CO., LTD.
(Manufacturer)

: 34-14 Maehama, Hiroishi-cho, Gamagori, Aichi 443-0038, JAPAN
Telephone: +81-533-67-6611
URL: <https://www.nidek.com/>

NIDEK INC.
(United States Agent)

: 2040 Corporate Court, San Jose, CA 95131, U.S.A.
Telephone: +1-800-223-9044 (USA Only)
URL: <https://usa.nidek.com/>

NIDEK S.A.
(EU Authorized Representative)

: Europarc, 13 rue Auguste Perret, 94042 Créteil, FRANCE
Telephone: +33 1 49 80 97 97
URL: <http://www.nidek.fr/>

2019-05
RT015-P9A2-A3
Printed in Japan

© 2018 NIDEK CO., LTD.

Before Use

This interface manual describes data communication between the Nidek RT-6100 and computers using wired LAN or wireless LAN. If you encounter any problems or have questions about the device, contact NIDEK or your authorized distributor.

See the following when the device is connected to the serial port of the computer:

- The optional memory box MEM-200 is required.
- Refer to the RT-5100 Interface Manual as well.

Table of Contents

1 CONNECTION WITH PERIPHERAL DEVICES - - - 3

- 1.1 System diagram example - - - 4

2 SETTINGS DEPENDING ON CONNECTION - - - 7

- 2.1 Data Output: RT-6100 to PC - - - 7
 - 2.1.1 Case 1 - Connection by LAN cable - - - 7
 - 2.1.2 Case 2 - Connection by LAN cable (output to computer via MEM-200 with serial communication) - - - 13
 - 2.1.3 Case 3 - Connection by wireless LAN - - - 16
 - 2.1.4 Case 4 - Connection by wireless LAN (output to computer via MEM-200 with serial communication) - - - 18
- 2.2 Data Input from Computer (Only for Data Transmission from Computer with Serial Communication) - - - 21
 - 2.2.1 Case 1 - Connection by LAN cable - - - 21
 - 2.2.2 Case 2 - Connection by wireless LAN - - - 24
- 2.3 Output of Drawing Data for Amsler Grid Test - - - 27

3 DATA FORMAT - - - 29

- 3.1 Data Output: RT-6100 to PC - - - 29
 - 3.1.1 XML file name - - - 29
 - 3.1.2 XML File Configuration of RT Data - - - 30
 - 3.1.3 XML File Sample - - - 46
 - 3.1.4 Serial data - - - 60
- 3.2 Data Input from Computer - - - 61
- 3.3 Output of Drawing Data for Amsler Grid Test - - - 62



CONNECTION WITH PERIPHERAL DEVICES

- This section describes the outline of LAN communication and examples of system configuration.

1

1.1 Outline of LAN Communication

- The RT-6100 creates XML (measurement results and information) and JPEG (image) files of the measurement data. It is equipped with a LAN interface that outputs the created files to the shared folders in external computers using CIFS (Common Internet File System).

* On Windows , enable SMB 1.0.

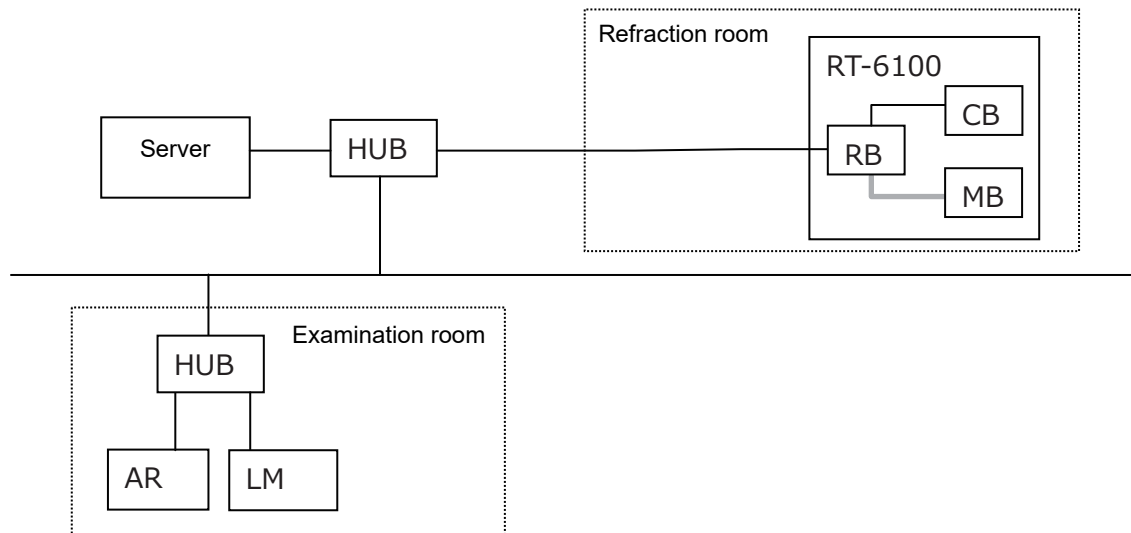
- Basic interface specifications

1	Basic specifications	CIFS
2	Connector	RJ-45
3	Standard (wired)	10BASE-T IEEE802.3/100BASE-TX IEEE802.3u
4	Standard (wireless)	IEEE802.11 a/b/g/n

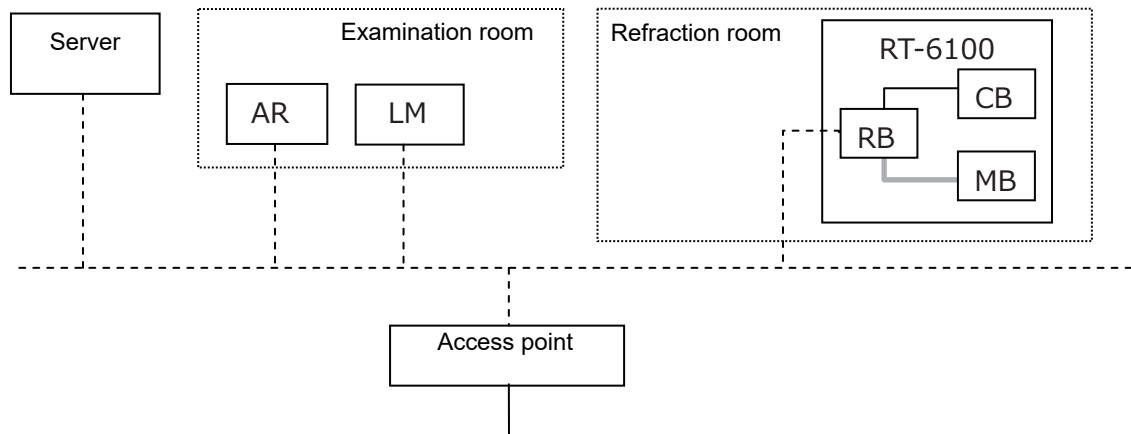
1.2 System diagram example

Data input: AR, LM → RT, Data output: RT → computer

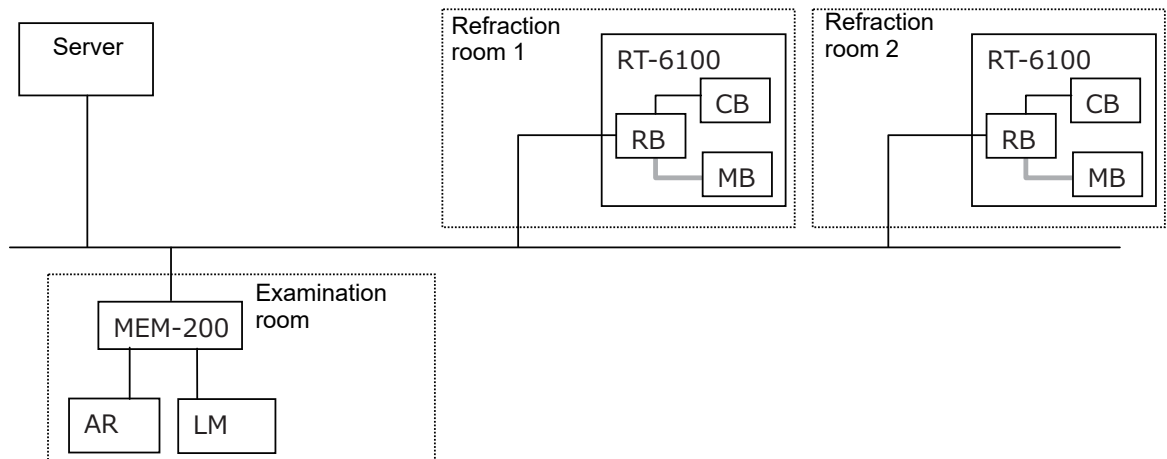
- When the shared folder is in RB: wired LAN



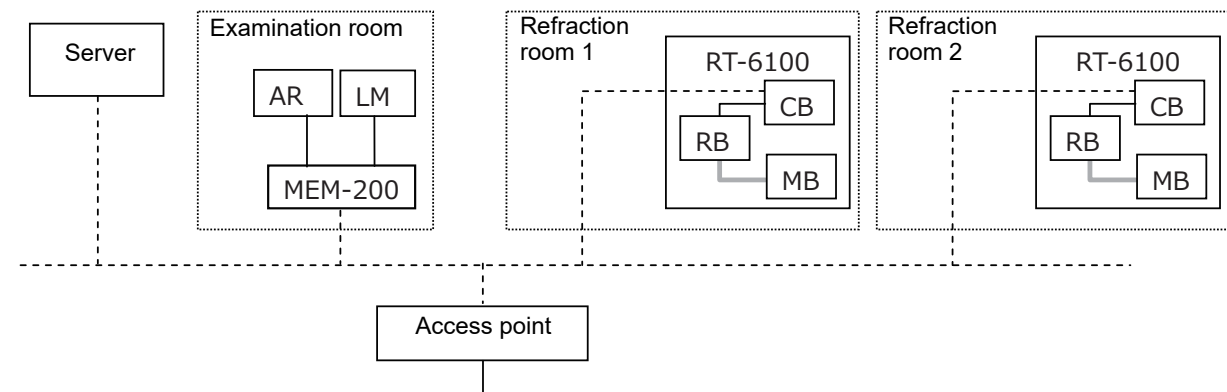
- When the shared folder is in RB: wireless LAN



- When the shared folder is in MEM-200: wired LAN



- When the shared folder is in MEM-200: wireless LAN



2

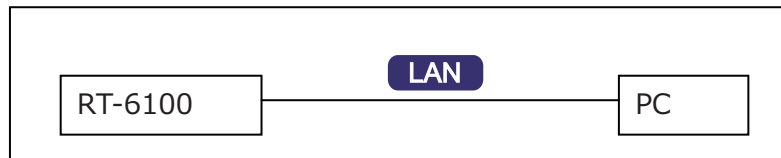
SETTINGS DEPENDING ON CONNECTION

- This section describes the connection diagram of peripheral devices, and the examples of the RT-6100 settings.
- Set the peripheral devices referring to the operator's manual and interface manual of each.

2.1 Data Output: RT-6100 to PC

2.1.1 Case 1 - Connection by LAN cable

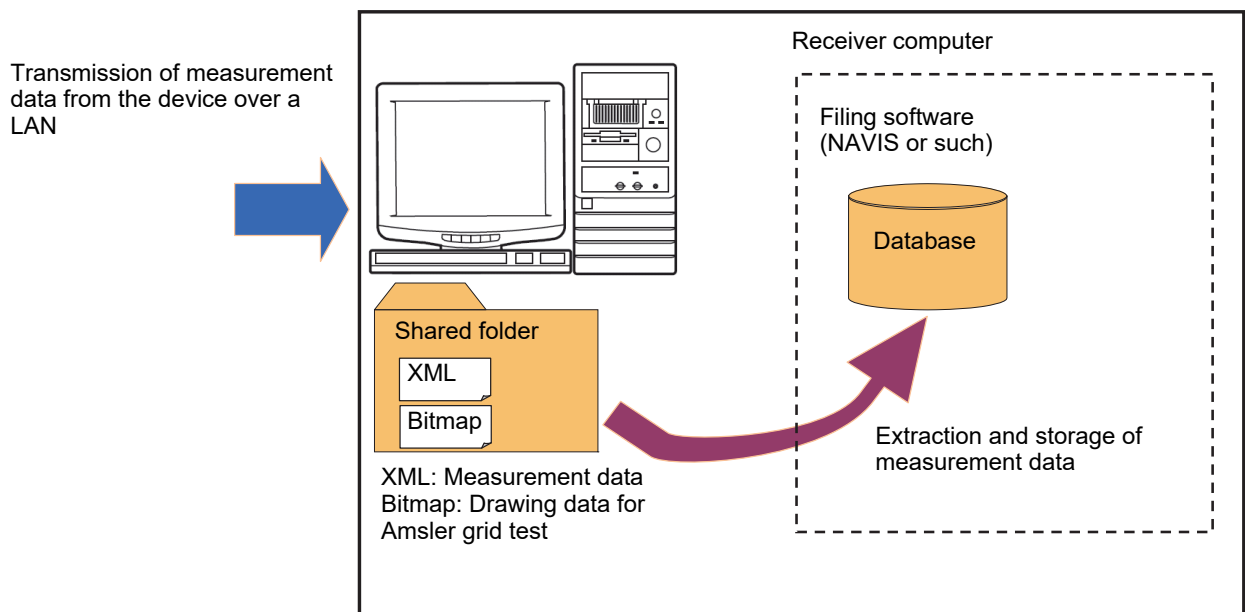
The LAN cable is connected to the LAN connector on the relay box and to the computer.



◆ Setting of receiver computer

- Reception structure of measurement data

The measurement data transmitted from the RT-6100 is output as shown in the figure below.



The measurement data transmitted by a LAN is saved in the shared folder in the receiver computer set with the RT-6100. The measurement data is saved in the shared folder in the XML format. If there are any drawing data for Amsler grid test, they are saved as bitmap files. After that, extract the measurement data using the filing software such as NAVIS, and save it in the database.

● Setting the shared folder

Set up a shared folder to save measurement data transmitted from the RT-6100.

Note

- Be sure that no shared folders have the same name.
The device cannot discern the folder and measurement data cannot be transmitted properly.
- * The name of the shared folder is the initial setting of the share name. (The share name is changeable.)

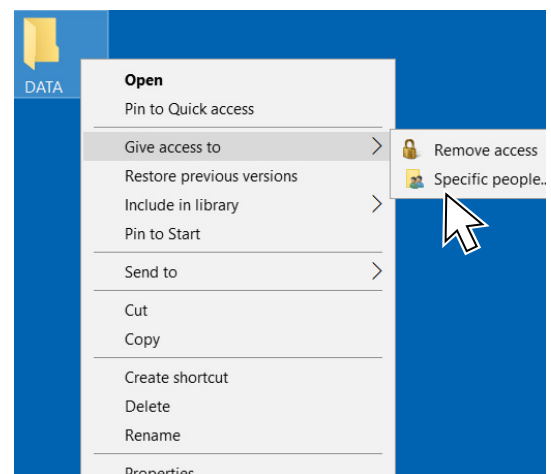
Setting a shared folder (folder name: DATA) to the desktop of the receiver computer

The procedure is explained using the example of Windows 10.

1) Newly create a folder on desktop and name the folder with "DATA".

2) Right-click the "DATA" folder, and select "Give access to" > "Specific people".

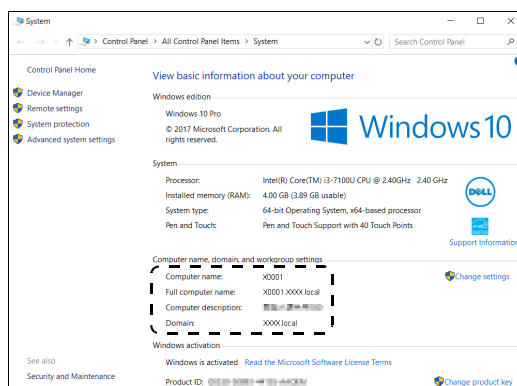
The network access window is displayed.



3) Confirm the following settings necessary for device connection.

(If necessary, confirm to the network administrator.)

- Name of users who can access the shared folder
- Password of users who can access the shared folder
- Name of the domain to which users who can access the shared folder belongs
(If no domain exists, confirm the work group name.)
- Computer name of the receiver computer (or IP address of the receiver computer)
(Confirm the computer name and domain using System Properties of the computer.)



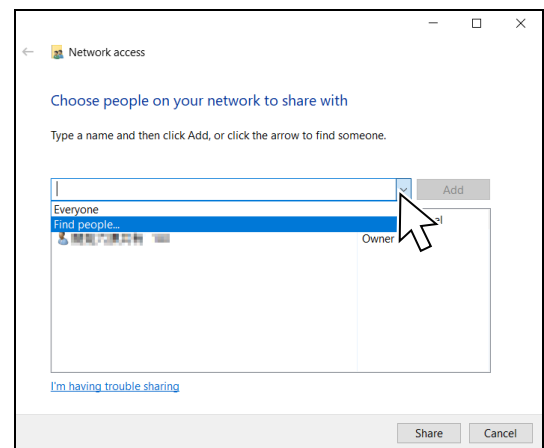
Example of Windows 10

- Shared folder name (set with "● Setting the shared folder" (page 8))
- IP address of the device (issued by the network administrator)
- Subnet mask of the device (issued by the network administrator)
- Default gateway (only when necessary)
- Availability of DHCP server (Confirm with network administrator.)

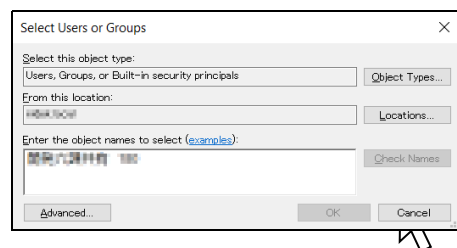
Ex.— Confirmation of each setting

Login user name	NIDEK
Login password	RT015
Domain name	WORKGROUP
Computer name (IP address of receiver computer)	192.168.0.1 (IP address of receiver computer)
Share name	DATA
IP address of the control box (wired LAN)	192.168.0.110
Subnet mask of the control box (wired LAN)	255. 255. 255. 0
IP address of the control box (wireless LAN)	192.168.0.111
Subnet mask of control box (wireless LAN)	255. 255. 255. 0
Default gateway (only when necessary)	0.0.0.0

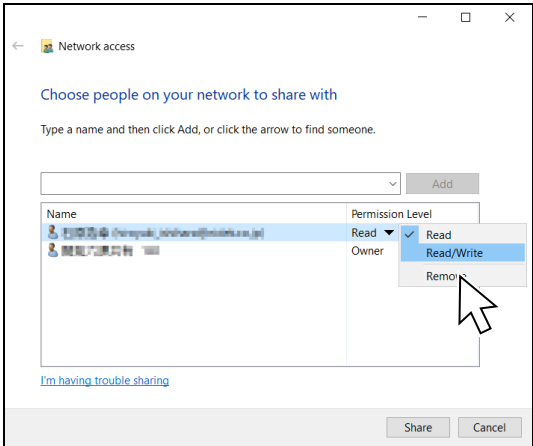
- 4) Press [V], and select “Find people...” from the menu.



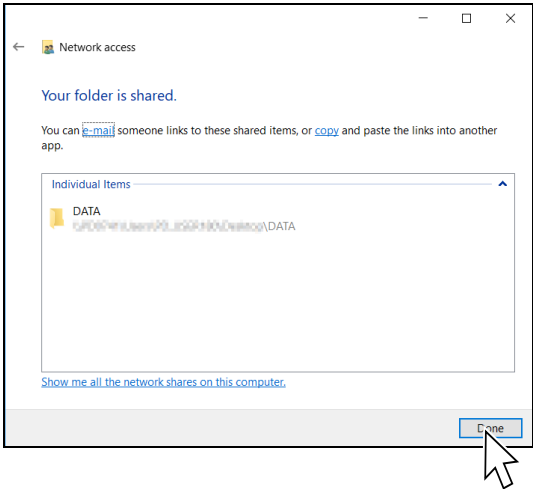
- 5) In the “Enter the object names to select” box, enter the name (or a part of it) of the user who that can access the shared folder, then click [OK].



- 6) Click the name of the user who can access the shared folder
- 7) Select “Read/Write”.
- 8) Click [Share].



- 9) Click [Done].



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	RB communication device setting	LAN
		File access communication device setting	LAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP

Major item	Secondary item	Setting item	Setting
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP
	Share	User name	NIDEK (changeable)
		Password	RT015 (changeable)
		Folder path	\DATA
		Share name	DATA
Other	External output	External output function	To be used
		Connection destination (computer name)	Depending on the setting of the destination computer
		Folder name	
		User name	
		Password	
		Domain name	
		Format	NIDEK_RT_V1.00

◆ Testing device network setting

After changing the network setting, turn off and on the device, and perform the operation below.

● Testing communication with the set shared folder

Press [Execution] in the item below to check each communication.

Communication success: ○, communication failure: ✕

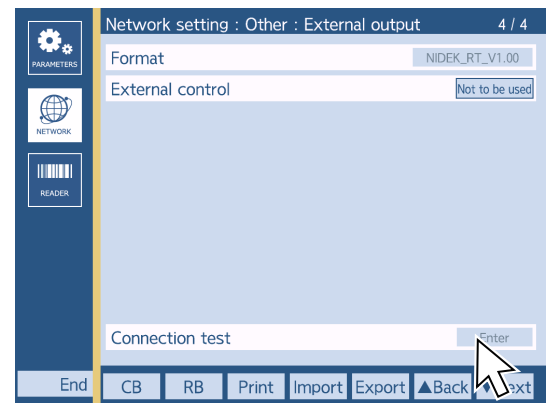
Secondary item	Setting item
Shared folder	Connection test

- Testing external output setting

Press [Execution] in the item below to check each communication.

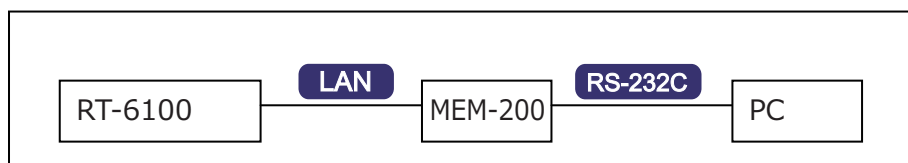
Communication success: ○, communication failure: ✕

Secondary item	Setting item
External output	Connection test



2.1.2 Case 2 - Connection by LAN cable (output to computer via MEM-200 with serial communication)

The LAN cable is connected to the LAN connector on the relay box and to the MEM-200. A serial communication is established between the MEM-200 and the computer.



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	RB communication device setting	LAN
		File access communication device setting	LAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP
Other	Shared folder	Shared folder location (computer name or IP address)	MEM-200 or 192.168.0.120
		Folder name	DATA
		Input folder type	MEM-200
		Input of serial communication computer data (MEM-200)	Port to which computer is connected

◆ Setting of receiver computer

Refer to the RT-5100 Interface Manual.

◆ MEM-200 setting

The network connection and the shared folder are set with the MEM-200. Refer to the operator's manual of the MEM-200.

		Setting item	Setting
Serial port setting	Serial port setting	PORT#2-#4	No.9
	Serial parameter	Communication sequence for PC	Type1 (Depending on the computer software setting)
Network setting	Shared folder settings	Maser PC name	MEM-200
		User name	nidek1 (default setting, changeable)
		Password	nidek1 (default setting, changeable)
		Domain/Workgroup	WORKGROUP
		Folder name	DATA
		Wired LAN settings Master IP address	192.168.0.120
	Wired LAN settings	DHCP	OFF
		Subnet mask	255.255.255.0
		Default gateway	0.0.0.0

● Paths to sub folders under shared folder in MEM-200

The xml file output to the folder from the RT-6100 is converted to serial data, then output to the computer connected to the specified port.

Paths to sub folders under shared folder in MEM-200	Serial port on MEM-200
DIRECT_PC_0A\TXT	Port#4 on master
DIRECT_PC_1A\TXT	Port#2 on slave 1
DIRECT_PC_1B\TXT	Port#4 on slave 1
DIRECT_PC_2A\TXT	Port#2 on slave 2
DIRECT_PC_2B\TXT	Port#4 on slave 2
DIRECT_PC_3A\TXT	Port#2 on slave 3
DIRECT_PC_3B\TXT	Port#4 on slave 3

◆ Testing the device network setting

Only tests related to the RT-6100 are described below. For the test related to MEM-200, refer to the operator's manual of the MEM-200.

After changing the network setting, turn off and on the device, and perform the operation below.

● Testing communication with the set shared folder

Press [Execution] in the item below to check each communication.

Communication success: ○, communication failure: ×

Secondary item	Setting item
Shared folder	Connection test

● Testing external output setting

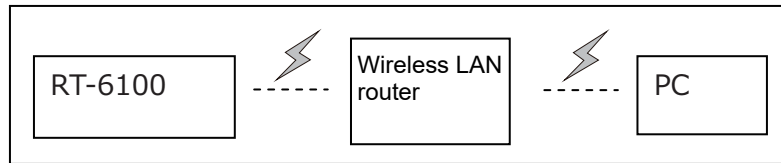
Press [Execution] in the item below to check each communication.

Communication success: ○, communication failure: ×

Secondary item	Setting item
External output	Connection test

2.1.3 Case 3 - Connection by wireless LAN

The settings of the computer, RT-6100, and wireless LAN router (when used) are described below.



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.111
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
		Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	RB communication device setting	LAN
		File access communication device setting	WLAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No
		Client→AP	Client

Major item	Secondary item	Setting item	Setting
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.113
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Client	Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP
	Share	User name	NIDEK (changeable)
		Password	RT015 (changeable)
		Folder path	\DATA
		Share name	DATA
Other	External output	External output function	To be used
		Connection destination (computer name)	Depending on the setting of the destination computer
		Folder name	
		User name	
		Password	
		Domain name	
		Format	NIDEK_RT_V1.00

◆ Setting of receiver computer

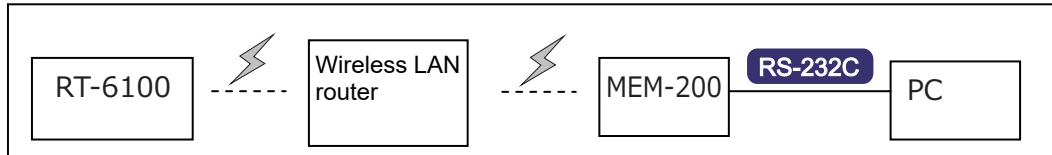
Refer to “◆ Setting of receiver computer” (page 7).

◆ Testing device network setting

Refer to “◆ Testing device network setting” (page 11).

2.1.4 Case 4 - Connection by wireless LAN (output to computer via MEM-200 with serial communication)

The settings of the RT-6100, MEM-200, and wireless LAN router (when used) are described below. A serial communication is established between the MEM-200 and the computer.



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.111
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
		Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	RB communication device setting	LAN
		File access communication device setting	WLAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No

Major item	Secondary item	Setting item	Setting
		Client→AP	Client
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.113
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Client	Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP
Other	Shared folder	Shared folder location (computer name or IP address)	MEM-200 or 192.168.0.120
		Folder name	DATA
		Input folder type	MEM-200
		Input of serial communication computer data (MEM-200)	Port to which computer is connected

◆ Setting of receiver computer

Refer to the RT-5100 Interface Manual.

◆ MEM-200 setting

The network connection and the shared folder are set with the MEM-200. Refer to the operator's manual of the MEM-200.

		Setting item	Setting
Serial port setting	Serial port setting	PORT#2-#4	No.9
	Serial parameter	Communication sequence for PC	Type1 (Depending on the computer software setting)
Network setting	Shared folder settings	Maser PC name	MEM-200
		User name	nidek1 (default setting, changeable)
		Password	nidek1 (default setting, changeable)
		Domain/Workgroup	WORKGROUP
		Folder name	DATA
		Wireless LAN settings Master IP address	192.168.0.124
	Wireless LAN settings	DHCP	OFF
		Subnet mask	255.255.255.0
		Default gateway	0.0.0.0
	Wireless parameter setting	Mode	Client
		Band	AUTO
		SSID	Enter in according to the connected access point.
		Security	
		PSK	

- Paths to sub folders under shared folder in MEM-200

Refer to “● Paths to sub folders under shared folder in MEM-200” (page 14).

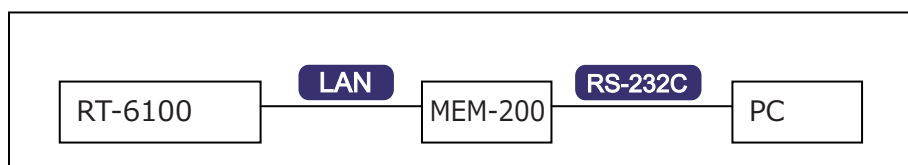
◆ Testing the device network setting

Refer to “◆ Testing device network setting” (page 17).

2.2 Data Input from Computer (Only for Data Transmission from Computer with Serial Communication)

2.2.1 Case 1 - Connection by LAN cable

The LAN cable is connected to the LAN connector on the relay box and to the MEM-200. A serial communication is established between the MEM-200 and the computer.



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	RB communication device setting	LAN
		File access communication device setting	LAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP

Major item	Secondary item	Setting item	Setting
Other	Shared folder	Shared folder location (computer name or IP address)	MEM-200 or 192.168.0.120
		Folder name	DATA
		Input folder type	MEM-200
		Input of serial communication computer data (MEM-200)	Port to which computer is connected

◆ Check and setting of computer

Refer to the RT-5100 Interface Manual.



Note

- If ID is not input when the data of RT-6100 is output, the ID information output with the serial communication is "NO_ID".

◆ MEM-200 setting

The network connection and the shared folder are set with the MEM-200. Refer to the operator's manual of the MEM-200.

		Setting item	Setting
Serial port setting	Serial port setting	PORT#2-#4	No.9
	Serial parameter	Communication sequence for PC	Type1 (Depending on the computer software setting)
Network setting	Shared folder settings	Maser PC name	MEM-200
		User name	nidek1 (default setting, changeable)
		Password	nidek1 (default setting, changeable)
		Domain/Workgroup	WORKGROUP
		Folder name	DATA
		Wired LAN settings Master IP address	192.168.0.120
	Wired LAN settings	DHCP	OFF
		Subnet mask	255.255.255.0
		Default gateway	0.0.0.0

- Paths to sub folders under shared folder in MEM-200

The serial data input from the computer connected to the specified port is converted to an xml file and provided to the RT-6100.

Paths to sub folders under shared folder in MEM-200	Serial port on MEM-200
DIRECT_RT_0A\TXT	Port#4 on master
DIRECT_RT_1A\TXT	Port#2 on slave 1
DIRECT_RT_1B\TXT	Port#4 on slave 1
DIRECT_RT_2A\TXT	Port#2 on slave 2
DIRECT_RT_2B\TXT	Port#4 on slave 2
DIRECT_RT_3A\TXT	Port#2 on slave 3
DIRECT_RT_3B\TXT	Port#4 on slave 3

2

◆ Testing device network setting

Only tests related to the RT-6100 are described below. For the test related to MEM-200, refer to the operator's manual of the MEM-200.

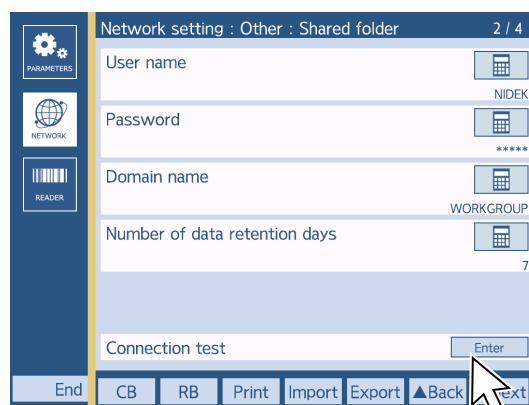
After changing the network setting, turn off and on the device, and perform the operation below.

- Testing communication with the set shared folder

Press [Execution] in the item below to check each communication.

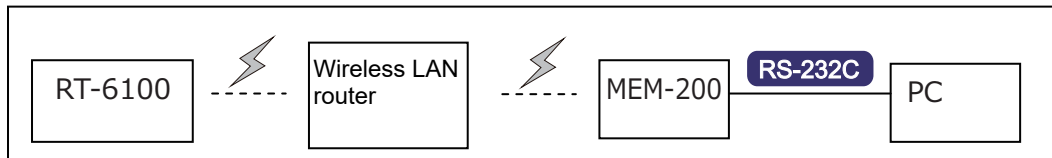
Communication success: ○, communication failure: ✕

Secondary item	Setting item
Shared folder	Connection test



2.2.2 Case 2 - Connection by wireless LAN

The settings of the RT-6100, MEM-200, and wireless LAN router (when used) are described below. A serial communication is established between the MEM-200 and the computer.



◆ RT-6100 settings

Major item	Secondary item	Setting item	Setting
CB	LAN	Use of DHCP	No
		IP address	192.168.0.110
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.111
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
		Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	RB communication device setting	LAN
		File access communication device setting	WLAN
		MB control communication port No.	59041
		RB communication port No.	59042
		External control communication port No.	62929
		Computer name	RT015_CB
		Domain	WORKGROUP
RB	LAN	Use of DHCP	No
		IP address	192.168.0.112
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	WLAN	Use of DHCP	No
		Client→AP	Client

Major item	Secondary item	Setting item	Setting
		WLAN radio wave strength	High
		IP address (IPv4)	192.168.0.113
		Subnet mask (IPv4)	255.255.255.0
		Default gateway (IPv4)	0.0.0.0
	Client	Frequency band used	Auto
		WLAN SSID	Enter in according to the connected access point.
		WLAN encryption method (security)	
		WLAN password	
	Basic information	MB control communication port No.	59041
		RB communication port No.	59042
		Computer name	RT015_RB
		Domain	WORKGROUP
Other	Shared folder	Shared folder location (computer name or IP address)	MEM-200 or 192.168.0.120
		Folder name	DATA
		Input folder type	MEM-200
		Input of serial communication computer data (MEM-200)	Port to which computer is connected

◆ Check and setting of computer

Refer to the RT-5100 Interface Manual.

◆ MEM-200 setting

The network connection and the shared folder are set with the MEM-200. Refer to the operator's manual of the MEM-200.

		Setting item	Setting
Network setting	Shared folder settings	Maser PC name	MEM-200
		User name	nidek1 (default setting, changeable)
		Password	nidek1 (default setting, changeable)
		Domain/Workgroup	WORKGROUP
		Folder name	DATA
		Wireless LAN settings Master IP address	192.168.0.124
	Wired LAN settings	DHCP	OFF
		Subnet mask	255.255.255.0
		Default gateway	0.0.0.0
	Wireless parameter setting	Mode	Client
		Band	AUTO
		SSID	Enter in according to the connected access point.
		Security	
		PSK	

- Paths to sub folders under shared folder in MEM-200

Refer to “● Paths to sub folders under shared folder in MEM-200” (page 23).

◆ Testing device network setting

Refer to “◆ Testing device network setting” (page 23).

2.3 Output of Drawing Data for Amsler Grid Test

The device network setting is tested in the same manner as “♦ *Testing device network setting*” (page 11).

DATA FORMAT

3.1.1 XML file name

RT_ABCDEFGHIJKLMNOPQRST_20170731_103057_01_.xml

a b c d e f

a	Header (2 characters fixed)
b	ID (Up to 20 characters)
c	Date (8 characters) (yyyymmdd)
d	Time (6 characters) (hhmmss)
e	System No. (2 characters) (Not displayed when System No. of RT is set to “Off”.)
f	None

3.1.2 XML File Configuration of RT Data

The tag configuration of an XML file is described below.

The tag format is NIDEK format.

In the actual data, there is no tag for data that was not output from the RT.

Tag name	Description
Common data <Common>	Common tag Version, measurement date and time, and such Patient data
Refractor data <Phoropter>	Measurement tag Refractor data
Unaided data <Unaided>	Unaided visual acuity data Far point, near point
Lensmeter (LM) data <Corrected>	Lensmeter refraction data Far point, near point
Objective (AR) data <Corrected>	Objective (AR) refraction data Far point
OPD data <Corrected>	OPD refraction data Far point
Subjective data <Corrected>	Subjective refraction data Far point, near point
Prescription data <Corrected>	Final prescription refraction data Far point, near point
Other data <Other>	Fusion data Stereoparallax data Aniseikonia data Dominant eye data Glare / Contrast visual acuity data ETDRS visual acuity data
21 Points Exams data <The21pointExams>	The 21 Point Exams in Optometry measurement data
AR measurement value <REF>	Objective accommodation data
KM measurement data <KM>	Corneal shape data
NT measurement value <TM>	Intraocular pressure data
Refraction time <Test_Time unit="sec">	Refraction time
System No. <SystemNo>	System No.

◆ Essential tags

Essential tag	Description
○	Data and tag are necessary.
△	Tag is necessary even if there is no data.
()	Tag is unnecessary if there is no data.

◆ Common tag definition

Tag name	Description	Data type	Necessity	Example
<Common>	Common tag	-	○	
<Company>	Company name	string	△	NIDEK
<ModelName>	Model name	string	○	RT-6100
<MachineNo>	Same model identification number	string	△	
<ROMVersion>	Device software version	string	△	
<Version>	XML specification version (fixed)	string	○	NIDEK_RT_V1.00
<Date>	Date code (YYYY-MM-DD)	date	○	2017-03-27
<Time>	Time (H/M/S)	time	○	12:25:23
<Patient>	Patient	-	○	
<No>	Patient No.	string	△	
<ID>	Patient ID (Up to 20 characters)	string	○	123456789012
<FirstName>	Patient's first name	string	△	
<MiddleName>	Patient's middle name	string	△	
<LastName>	Patient's last name	string	△	
<Sex>	Patient's sex	string	△	
<Age>	Patient's age	int	△	25
<DOB>	Patient's date of birth	date	△	
<NameJ1>	Multilingual patient's name	string	△	
<NameJ2>	Multilingual patient's name	string	△	
<Operator>	Operator	-	()	
<No>	Operator No.	string	△	
<ID>	Operator ID	string	△	

❖ The maximum size of string is 64 bytes.

- "NO_ID" is indicated if there is no ID.
- The grayed out tags above are empty tags.

◆ Measurement tag definition

Type

Type=	Model identification
"RT"	Refractor

Tag name	Description	Data type	Necessity	Example
<Measure Type="RT" >	Measurement tag	-	○	
<Phoropter>	Phoropter	-	○	
<DiopterStep unit="D">	Diopter unit	double	△	
<AxisStep unit="deg">	Angle unit	int	△	
<CylinderMode>	Cylinder mode " + " / " - "	string	△	-
<Unaided>	Unaided eye measurement	-	()	Reference 1
<Corrected>	Correction measurement · LM data · Objective (AR) data · OPD data · Subjective data · Final prescription data	-	()	Reference 2
<Other>	Other measurements	-	()	Reference 3
<The21pointExams>	The 21 Point Exams in Optometry measurements	-	()	Reference 4
<REF>	Objective measurement	-	()	Reference 5
<KM>	Corneal shape measurement	-	()	Reference 6
<TM>	Intraocular pressure measurement	-	()	Reference 7
<Test_Time unit="sec">	Refraction time	int	()	201
<SystemNo>	System No.	int	△	1

❖ The maximum size of string is 64 bytes.

- The grayed out tags above are empty tags.

◆ Unaided tag definition (Reference 1)

Vision

Vision =	Refraction distance
"Distant"	Far point
"Near"	Near point

Situation

Situation =	Examination conditions
"Standard"	Day time, or no specification of either day time or night time
"Night"	Night time

Tag name	Description	Data type	Necessity	Example
<Unaided Vision="Distant" Situation="Standard">	Unaided eye measurement	-	()	
<WorkingDistance unit="cm">	Working distance (ADD)	int	△	
<R> or <L> or 	Right eye, left eye, or both eyes	-	()	
<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1
<PHVA>	Visual acuity (Pin hole)	string	()	>500 0.05 – 1

❖ The maximum size of string is 64 bytes.

- The grayed out tags above are empty tags.
- The visual acuity data may be expressed in formats such as decimal or fraction notation.

◆ Corrected tag definition (Reference 2)

CorrectionType

CorrectionType =	Explanation of correction procedure
"LM_Base"	LM data
"REF_Base"	Objective (AR) data
"WF_Base"	OPD data
"Full"	Subjective data
"Best"	Prescription data

Vision

Vision =	Refraction distance
"Distant"	Far point
"Near"	Near point

Situation

Situation =	Examination conditions
"Standard"	Day time, or no specification of either day time or night time
"Night"	Night time

Tag name	Description	Data type	Necessity	Example
<Corrected CorrectionType="REF_Base" Vision="Distant" Situation="Standard">	Corrected measurement tag	-	()	
<DisplayName>	Term used for display	string	△	
<VD unit="mm">	Vertex distance	double	△	
<WorkingDistance unit="cm">	Working distance (ADD)	int	△	
<WorkingDistance2 unit="cm">	Working distance (ADD2)	int	()	
<R> or <L>	Right eye or left eye	-	()	
<Sphere unit="D">	Spherical power	double	○	-1.25 0.00 +2.50
<Cylinder unit="D">	Refractive cylinder	double	○	-0.75 0.00 +1.25
<Axis unit="deg">	Cylindrical axis	int	○	179
<ADD unit="D">	Addition power	double	△	0.00 +2.00 -1.50
<ADD2 unit="D">	Addition power 2	double	()	
<Prism unit="pri">	Prism power	double	△	0.50
<PrismBase unit="deg">	Prism base angle	int	△	180

Tag name		Description	Data type	Necessity	Example
	<PrismX unit="pri" base="in" or "out" or "">	Horizontal prism power	double	△	"in" 2.50 "out" 1.50 "" 0.00
	<PrismY unit="pri" base="up" or "down" or "">	Vertical prism power	double	△	"up" 2.50 "down" 1.50 "" 0.00
	<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1
	<ADDVA>	Visual acuity (ADD)	string	△	
	<ADD2VA>	Visual acuity (ADD2)	string	()	
	<PHVA>	Visual acuity (Pin hole)	string	()	<0.04 0.05 + 1 >500 500 + 1
	<LIVA>	Visual acuity (Night chart)	string	()	
	<PD unit="mm">	Pupillary distance	double	△	31.5
		Both eyes	-	()	
	<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1
	<ADDVA>	Visual acuity (ADD)	string	△	
	<ADD2VA>	Visual acuity (ADD2)	string	()	
	<PHVA>	Visual acuity (Pin hole)	string	()	<0.04 0.05 + 1 >500 500 + 1
	<LIVA>	Visual acuity (Night chart)	string	()	
	<PD unit="mm">	Pupillary distance	double	△	63.0
	<Div>	Divergence	-	()	
	<Blur unit="pri">	Blur (Prism)	double	△	4.00 -2.40
	<Break unit="pri">	Break (Prism)	double	△	
	<Recovery unit="pri">	Recovery (Prism)	double	△	+2.40 +0.00 -1.20
	<Conv>	Convergence	-	()	
	<Blur unit="pri">	Blur (Prism)	double	△	4.00 -2.40
	<Break unit="pri">	Break (Prism)	double	△	
	<Recovery unit="pri">	Recovery (Prism)	double	△	+2.40 +0.00 -1.20
	<Supra>	Supravergence	-	()	
	<RL>	Prism added eye	string	○	R
	<Break unit="pri">	Break (Prism)	double	○	1.50 0.00 -1.20
	<Recovery unit="pri">	Recovery (Prism)	double	○	

Tag name			Description	Data type	Necessity	Example
		<Infra>	Infravergence	-	()	
		<RL>	Prism added eye	string	○	R
		<Break unit="pri">	Break (Prism)	double	○	1.50 0.00
		<Recovery unit="pri">	Recovery (Prism)	double	○	-1.20

❖ The maximum size of string is 64 bytes.

- The grayed out tags above are empty tags.
- The visual acuity data may be expressed in formats such as decimal or fraction notation.

◆ Tag definition of “Other” (Reference 3)

Tag name	Description	Data type	Necessity	Example
<Other>	Other tests	-	()	
<Fusion>	Fusion data “2”/“3”/“4”/“5”	string	()	4
<Stereo>	Stereoparallax 10'00": “1000” 8": “ 8X”	string	()	1000
<LIStereo>	Stereoparallax (Night chart)	string	()	8X
<Aniseikonia>	Aniseikonia	-	()	
<Horizontal>	Horizontal “00”/“99”	string	()	00
<Vertical>	Vertical “00”/“99”	string	()	99
<Dominant_Eye>	Dominant eye “R”/“L”	string	()	R
<Glare_Contrast>	Glare, contrast	-	()	
<Contrast>	Contrast level “ 0” to “100”/“000”	string	○	25
<Glare>	Glare level “H”/“M”/“L”/“N”	string	○	H
<R> or <L>	Right eye or left eye	-	()	
<Sphere unit=“D”>	Spherical power	double	○	-3.00
<Cylinder unit=“D”>	Refractive cylinder	double	○	-1.00
<Axis unit=“deg”>	Cylindrical axis	int	○	180
<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1
	Both eyes	-	()	
<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1
<ETDRS>	ETDRS	-	()	
<Contrast>	Contrast level “ 0” to “100”/“000”	string	○	25
<Glare>	Glare level “H”/“M”/“L”/“N”	string	○	H
<R> or <L>	Right eye or left eye	-	()	
<Sphere unit=“D”>	Spherical power	double	○	-3.00
<Cylinder unit=“D”>	Refractive cylinder	double	○	-1.00
<Axis unit=“deg”>	Cylindrical axis	int	○	180
<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1

Tag name			Description	Data type	Necessity	Example
			Both eyes	-	()	
		<VA>	Visual acuity	string	△	<0.04 0.05 + 1 >500 500 + 1

❖ The maximum size of string is 64 bytes.

- The visual acuity data may be expressed in formats such as decimal or fraction notation.

◆ Tag definition of “The21pointExams” (Reference 4)

Tag name	Description	Data type	Necessity	Example
<The21pointExams>	The 21 Point Exams in Optometry		()	
<No07>	Full correction value Same as subjective (far) day data		()	
<R> or <L>	Right eye or left eye	-	()	
<Sphere unit="D">	Spherical power	double	○	-3.00
<Cylinder unit="D">	Refractive cylinder	double	○	-1.00
<Axis unit="deg">	Cylindrical axis	int	○	180
<No0910>	Convergence (Far) Same as subjective (far) day data		()	
<Blur unit="pri">	Blur (Prism)	double	○	4.00
<Break unit="pri">	Break (Prism)	double	○	-2.40
<Recovery unit="pri">	Recovery (Prism)	double	○	+2.40 +0.00 -1.20
<No11>	Divergence (Far) Same as subjective (far) day data		()	
<Blur unit="pri">	Blur (Prism)	double	○	4.00
<Break unit="pri">	Break (Prism)	double	○	-2.40
<Recovery unit="pri">	Recovery (Prism)	double	○	+2.40 +0.00 -1.20
<No12B>	Supravergence/infravergence (Far) Same as subjective (far) day data		()	
<Supra>	Supravergence		()	
<RL>	Prism added eye	string	○	R
<Break unit="pri">	Break (Prism)	double	○	1.50
<Recovery unit="pri">	Recovery (Prism)	double	○	0.00 -1.20
<Infra>	Infravergence	double	()	
<RL>	Prism added eye	string	○	R
<Break unit="pri">	Break (Prism)	double	○	1.50
<Recovery unit="pri">	Recovery (Prism)	double	○	0.00 -1.20
<No14A>	Cross cylinder (Monocular accommodative lag) (Only Plus Package)		()	
<R> or <L>	Right eye or left eye	-	()	
<ADD unit="D">	Addition power	double	○	-1.00

Tag name	Description	Data type	Necessity	Example
<No14B>	Cross cylinder (Binocular accommodative lag) (Only Plus Package)		()	
<R> or <L>	Right eye or left eye	-	()	
<ADD unit="D">	Addition power	double	○	-1.00
<No15A>	Horizontal phoria by #14A (Only Plus Package)		()	
<PrismH unit="pri" phoria="Eso" or "Exo" or "">	Horizontal prism power	double	○	"Eso" 0.50 "Exo" 1.50 "" 0.00
<No15B>	Horizontal phoria by #14B (Only Plus Package)		()	
<PrismH unit="pri" phoria="Eso" or "Exo" or "">	Horizontal prism power	double	○	"Eso" 0.50 "Exo" 1.50 "" 0.00
<No16>	Convergence (Near) Same as subjective (near) day data		()	
<Blur unit="pri">	Blur (Prism)	double	○	4.00
<Break unit="pri">	Break (Prism)	double	○	-2.40
<Recovery unit="pri">	Recovery (Prism)	double	○	+2.40 +0.00 -1.20
<No17>	Divergence (Near) Same as subjective (near) day data		()	
<Blur unit="pri">	Blur (Prism)	double	○	4.00
<Break unit="pri">	Break (Prism)	double	○	-2.40
<Recovery unit="pri">	Recovery (Prism)	double	○	+2.40 +0.00 -1.20
<No18B>	Supravergence/infravergence (Near) Same as subjective (near) day data		()	
<Supra>	Supravergence		()	
<RL>	Prism added eye	string	○	R
<Break unit="pri">	Break (Prism)	double	○	1.50
<Recovery unit="pri">	Recovery (Prism)	double	○	0.00 -1.20
<Infra>	Infravergence		()	
<RL>	Prism added eye	string	○	R
<Break unit="pri">	Break (Prism)	double	○	1.50
<Recovery unit="pri">	Recovery (Prism)	double	○	0.00 -1.20
<No19>	Accommodation		()	

Tag name	Description	Data type	Necessity	Example
<MinusLens>			()	
<R> or <L> or 	Right eye, left eye, or both eyes	-	()	
<ACC unit="D">	Accommodation	double	○	+2.75
<NPA>			()	
<R> or <L> or 	Right eye, left eye, or both eyes	-	()	
<ACC unit="D">	Accommodation	double	○	3.26
<Distance unit="cm">		double	△	21
<No20>	PRA (Positive relative accommodation)		()	
<R> or <L> or 	Right eye, left eye, or both eyes	-	()	
<Blur unit="D">	Blur (D)	double	○	-1.50
<Recovery unit="D">	Recovery (D)	double	○	-0.00
<No21>	NRA (Negative relative accommodation)		()	
<R> or <L> or 	Right eye, left eye, or both eyes	-	()	
<Blur unit="D">	Blur (D)	double	○	+2.50
<Recovery unit="D">	Recovery (D)	double	○	+0.00
<NPC>	NPC (Near point of convergence)		()	
<Distance unit="cm">	Distance (cm)	int	○	45
<MA unit="MA">	Meter angle (MA)	double	○	2.1
<Prism unit="pri">	Prism (pri)	double	○	13.8
<GradACA>	Gradient AC/A Not coexistent with Heterophoria, except AC/A (Only Plus Package)		()	
<ACA>	AC/A		()	4.00
<Sheard>	Sheard's criterion		()	
<Distant>	Far point		()	
<PrismX unit="pri" base="in" or "out" or "">	Prism	double	△	"in" 2.50 "out" 1.50 "" 0.00
<Sphere unit="D">	Spherical power	double	△	+0.25 0.00
<Near>	Near point		()	
<PrismX unit="pri" base="in" or "out" or "">	Prism	double	△	"in" 2.50 "out" 1.50 "" 0.00
<Sphere unit="D">	Spherical power	double	△	-0.25 0.00

Tag name	Description	Data type	Necessity	Example
<Percival>	Percival's criterion		()	
<Distant>	Far point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	-0.25 0.00
<Near>	Near point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	+0.25 0.00
<HeteroACA>	Heterophoria AC/A Not coexistent with Gradient, except AC/A (Only Plus Package)		()	
<ACA>	AC/A		()	4.00
<Sheard>	Sheard's criterion		()	
<Distant>	Far point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	+0.25 0.00
<Near>	Near point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	-0.25 0.00
<Percival>	Percival's criterion		()	
<Distant>	Far point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	-0.25 0.00
<Near>	Near point		()	
<PrismX unit="pri" base="in" or "out" or ">	Prism	double	△	"in" 2.50 "out" 1.50 " " 0.00
<Sphere unit="D">	Spherical power	double	△	+0.25 0.00

Tag name	Description	Data type	Necessity	Example
<Morgan>	Morgan analysis (Only Plus Package)		()	
<A>	Analysis results of Group A Item 5	string	△	HHHHH
	Analysis results of Group B Item 7	string	△	NNNNNNN
<C>	Analysis results of Group C Item 3	string	△	LLL

❖ The maximum size of string is 64 bytes.

- The visual acuity data may be expressed in formats such as decimal or fraction notation.

◆ Tag definition of “REF” (Reference 5)

Tag name	Description	Data type	Necessity	Example
<REF>	AR measurement data			
<ReferencedInstance>	Reference file	string	△	
<DiopterStep unit="D">	Diopter unit	double	△	
<AxisStep unit="deg">	Angle unit	int	△	
<VD unit="mm">	Vertex distance	int	△	
<CylinderMode>	Cylinder mode “+”/“-”/“MIX”	string	△	-
<Objective_Acc>	Objective accommodation		()	
<R> or <L>	Right eye or left eye	-	()	
<ACC unit="D">	Accommodation	double	○	3.00

- ❖ The maximum size of string is 64 bytes.
- The grayed out tags above are empty tags.

◆ Tag definition of “KM” (Reference 6)

Tag name	Description	Data type	Necessity	Example
<KM>	KM measurement data		()	
<ReferencedInstance>	Reference file	string	△	
<DiopterStep unit="D">	Diopter unit	double	△	
<AxisStep unit="deg">	Angle unit	int	△	
<CylinderMode>	Cylinder mode “+”/“-”	string	△	
<R> or <L>	Right eye or left eye	-	()	
<Median>	Median values		○	
<R1>	Flattest meridian		○	
<Radius unit="mm">	Curvature radius	double	○	7.97
<Power unit="D">	Corneal refractive power	double	○	42.38
<Axis unit="deg">	Cylindrical axis	int	○	180
<R2>	Steepest meridian		○	
<Radius unit="mm">	Curvature radius	double	○	7.96
<Power unit="D">	Corneal refractive power	double	○	42.50
<Axis unit="deg">	Cylindrical axis	int	△	90

- ❖ The maximum size of string is 64 bytes.
- The grayed out tags above are empty tags.

◆ Tag definition of “TM” (Reference 7)

Tag name	Description	Data type	Necessity	Example
<TM>	NT measurement data		()	
<ReferencedInstance>	Reference file	string	△	
<R> or <L>	Right eye or left eye	-	()	
<Average>			○	
<IOP_mmHg unit="mmHg">	Intraocular pressure value (mmHg)	double	○	12.5
<IOP_Pa unit="kPa">	Intraocular pressure value (kPa)	double	△	1.3

- ❖ The maximum size of string is 64 bytes.
- The grayed out tags above are empty tags.

3.1.3 XML File Sample

- This is not a collection of actual measurement data but is only a sample for explanation.

```
<?xml version="1.0" encoding="UTF-16"?>
<?xml-stylesheet type="text/xsl" href="NIDEK_RT_Stylesheet.xml"?>
<Ophthalmology>
  <Common>
    <Company>NIDEK</Company>
    <ModelName>RT-6100</ModelName>
    <MachineNo></MachineNo>
    <ROMVersion></ROMVersion>
    <Version>NIDEK_RT_V1.00</Version>
    <Date>2017-05-28</Date>
    <Time>11:29:49</Time>
    <Patient>
      <No></No>
      <ID>112233445501</ID>
      <FirstName></FirstName>
      <MiddleName></MiddleName>
      <LastName></LastName>
      <Sex></Sex>
      <Age>23</Age>
      <DOB></DOB>
      <NameJ1></NameJ1>
      <NameJ2></NameJ2>
    </Patient>
  </Common>
  <Measure Type="RT">
    <Phoropter>
      <DiopterStep unit="D"></DiopterStep>
      <AxisStep unit="deg"></AxisStep>
      <CylinderMode></CylinderMode>
      <Unaided Vision="Distant" Situation="Standard">
        <WorkingDistance unit="cm">40</WorkingDistance>
        <R>
          <VA> 0.04-1</VA>
          <PHVA>&lt;0.04 </PHVA>
        </R>
        <L>
          <VA> 0.04+2</VA>
          <PHVA> 0.04+1</PHVA>
        </L>
        <B>
          <VA>&lt;0.04 </VA>
          <PHVA> 0.04 -1</PHVA>
        </B>
      </Unaided>
      <Unaided Vision="Near" Situation="Standard">
        <WorkingDistance unit="cm">40</WorkingDistance>
        <R>
          <VA> 0.04+1</VA>
          <PHVA> 0.04+1</PHVA>
        </R>
        <L>
          <VA> 0.04 +1</VA>
          <PHVA>&lt;0.04 </PHVA>
        </L>
        <B>
```

```

    <VA>&lt;0.04 </VA>
    <PHVA> 0.04 -1</PHVA>
  </B>
</Unaided>
<Unaided Vision="Distant" Situation="Night">
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <VA> 0.05 +1</VA>
    <PHVA> 0.04 -1</PHVA>
  </R>
  <L>
    <VA> 0.04-1</VA>
    <PHVA> 0.04+1</PHVA>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <PHVA>&lt;0.04 </PHVA>
  </B>
</Unaided>
<Corrected CorrectionType="LM_Base" Vision="Distant" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">31</WorkingDistance>
  <R>
    <Sphere unit="D">-7.25</Sphere>
    <Cylinder unit="D">-2.00</Cylinder>
    <Axis unit="deg">169</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="up">11.00</PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA> 0.04+1</ADDVA>
    <PHVA> 0.04+1</PHVA>
    <LIVA> 0.04-1</LIVA>
    <PD unit="mm">32.5</PD>
  </R>
  <L>
    <Sphere unit="D">-7.50</Sphere>
    <Cylinder unit="D">-0.25</Cylinder>
    <Axis unit="deg">149</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="down">11.00</PrismY>
    <VA> 0.04-1</VA>
    <ADDVA> 0.04 -1</ADDVA>
    <PHVA> 0.04 -1</PHVA>
    <LIVA> 0.04+1</LIVA>
    <PD unit="mm">33.0</PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA>&lt;0.04 </ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <LIVA>&lt;0.04 </LIVA>
    <PD unit="mm">65.5</PD>

```

```

</B>
</Corrected>
<Corrected CorrectionType="LM_Base" Vision="Distant" Situation="Night">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">31</WorkingDistance>
  <R>
    <Sphere unit="D">-7.25</Sphere>
    <Cylinder unit="D">-2.00</Cylinder>
    <Axis unit="deg">169</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="out">0.50</PrismX>
    <PrismY unit="pri" base="down">11.50</PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA> 0.04+1</ADDVA>
    <PHVA> 2.0 -1</PHVA>
    <PD unit="mm"></PD>
  </R>
  <L>
    <Sphere unit="D">-7.50</Sphere>
    <Cylinder unit="D">-0.25</Cylinder>
    <Axis unit="deg">149</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="out">0.50</PrismX>
    <PrismY unit="pri" base="up">11.50</PrismY>
    <VA> 0.04-1</VA>
    <ADDVA> 0.04 -1</ADDVA>
    <PHVA> 2.0 +1</PHVA>
    <PD unit="mm"></PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA>&lt;0.04 </ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <PD unit="mm"></PD>
  </B>
</Corrected>
<Corrected CorrectionType="REF_Base" Vision="Distant" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm"></WorkingDistance>
  <R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D"></ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base=""></PrismX>
    <PrismY unit="pri" base=""></PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA></ADDVA>
    <PHVA> 0.04 -1</PHVA>
    <LIVA>&lt;0.04 </LIVA>
    <PD unit="mm">32.5</PD>

```

```

</R>
<L>
  <Sphere unit="D">-7.75</Sphere>
  <Cylinder unit="D">-0.75</Cylinder>
  <Axis unit="deg">30</Axis>
  <ADD unit="D"></ADD>
  <Prism unit="pri"></Prism>
  <PrismBase unit="deg"></PrismBase>
  <PrismX unit="pri" base=""></PrismX>
  <PrismY unit="pri" base=""></PrismY>
  <VA> 0.04-1</VA>
  <ADDVA></ADDVA>
  <PHVA>&lt;0.04 </PHVA>
  <LIVA> 0.04-1</LIVA>
  <PD unit="mm">33.0</PD>
</L>
<B>
  <VA>&lt;0.04 </VA>
  <ADDVA></ADDVA>
  <PHVA> 0.04+1</PHVA>
  <LIVA> 0.04+1</LIVA>
  <PD unit="mm">65.5</PD>
</B>
</Corrected>
<Corrected CorrectionType="REF_Base" Vision="Distant" Situation="Night">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm"></WorkingDistance>
  <R>
    <Sphere unit="D">-8.50</Sphere>
    <Cylinder unit="D">-1.25</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D"></ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base=""></PrismX>
    <PrismY unit="pri" base=""></PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA></ADDVA>
    <PHVA> 0.04 -1</PHVA>
    <PD unit="mm">32.5</PD>
  </R>
  <L>
    <Sphere unit="D">-8.00</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
    <ADD unit="D"></ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base=""></PrismX>
    <PrismY unit="pri" base=""></PrismY>
    <VA> 0.04-1</VA>
    <ADDVA></ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <PD unit="mm">33.0</PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA></ADDVA>

```

```

    <PHVA> 0.04+1</PHVA>
    <PD unit="mm">65.5</PD>
  </B>
</Corrected>
<Corrected CorrectionType="Full" Vision="Distant" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="out">0.50</PrismX>
    <PrismY unit="pri" base="down">0.50</PrismY>
    <VA> 2.0 +1</VA>
    <ADDVA> 2.0 -1</ADDVA>
    <PHVA> 2.0 +1</PHVA>
    <LIVA> 0.04+1</LIVA>
    <PD unit="mm">32.5</PD>
  </R>
  <L>
    <Sphere unit="D">-7.75</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="out">0.50</PrismX>
    <PrismY unit="pri" base="up">0.50</PrismY>
    <VA> 2.0 -1</VA>
    <ADDVA> 0.04+1</ADDVA>
    <PHVA> 0.04+1</PHVA>
    <LIVA> 0.04-1</LIVA>
    <PD unit="mm">33.0</PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA>&lt;0.04 </ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <LIVA>&lt;0.04 </LIVA>
    <PD unit="mm">65.5</PD>
    <Div>
      <Blur unit="pri">-0.60</Blur>
      <Break unit="pri">-1.20</Break>
      <Recovery unit="pri">-1.80</Recovery>
    </Div>
    <Conv>
      <Blur unit="pri">0.40</Blur>
      <Break unit="pri">0.80</Break>
      <Recovery unit="pri">+1.20</Recovery>
    </Conv>
    <Supra>
      <RL>R</RL>
      <Break unit="pri">0.60</Break>
      <Recovery unit="pri">-0.20</Recovery>
    </Supra>
  </B>

```

```

<Infra>
  <RL>R</RL>
  <Break unit="pri">0.40</Break>
  <Recovery unit="pri">-0.60</Recovery>
</Infra>
</B>
</Corrected>
<Corrected CorrectionType="Full" Vision="Near" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <Sphere unit="D">-6.75</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D"></ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="up">12.00</PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA></ADDVA>
    <PHVA> 0.04 -1</PHVA>
    <PD unit="mm">30.5</PD>
  </R>
  <L>
    <Sphere unit="D">-6.25</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
    <ADD unit="D"></ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="down">12.00</PrismY>
    <VA> 2.0 -1</VA>
    <ADDVA></ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <PD unit="mm">31.0</PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA></ADDVA>
    <PHVA> 0.04+1</PHVA>
    <PD unit="mm">61.5</PD>
    <Div>
      <Blur unit="pri">0.40</Blur>
      <Break unit="pri">0.80</Break>
      <Recovery unit="pri">+1.20</Recovery>
    </Div>
    <Conv>
      <Blur unit="pri">-0.40</Blur>
      <Break unit="pri">-0.80</Break>
      <Recovery unit="pri">-1.20</Recovery>
    </Conv>
  <Supra>
    <RL>R</RL>
    <Break unit="pri">0.40</Break>
    <Recovery unit="pri">0.90</Recovery>
  </Supra>

```

```

<Infra>
  <RL>R</RL>
  <Break unit="pri">0.80</Break>
  <Recovery unit="pri">1.40</Recovery>
</Infra>
</B>
</Corrected>
<Corrected CorrectionType="Full" Vision="Distant" Situation="Night">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <Sphere unit="D">-8.50</Sphere>
    <Cylinder unit="D">-1.25</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="up">0.50</PrismY>
    <VA> 0.04 +1</VA>
    <ADDVA> 0.04+1</ADDVA>
    <PHVA> 2.0 +1</PHVA>
    <PD unit="mm">32.5</PD>
  </R>
  <L>
    <Sphere unit="D">-8.00</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">0.50</PrismX>
    <PrismY unit="pri" base="down">0.50</PrismY>
    <VA> 0.04-1</VA>
    <ADDVA> 2.0 -1</ADDVA>
    <PHVA> 2.0 +1</PHVA>
    <PD unit="mm">33.0</PD>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
    <ADDVA> 2.0 </ADDVA>
    <PHVA>&lt;0.04 </PHVA>
    <PD unit="mm">65.5</PD>
  </B>
</Corrected>
<Corrected CorrectionType="Best" Vision="Distant" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D">+1.50</ADD>
    <Prism unit="pri"></Prism>
    <PrismBase unit="deg"></PrismBase>
    <PrismX unit="pri" base="in">1.00</PrismX>
    <PrismY unit="pri" base="up">1.50</PrismY>

```

```

<VA> 0.04 +1</VA>
<ADDVA> 0.04+1</ADDVA>
<PHVA> 0.04 -1</PHVA>
<LIVA>&lt;0.04 </LIVA>
<PD unit="mm">32.5</PD>
</R>
<L>
  <Sphere unit="D">-7.75</Sphere>
  <Cylinder unit="D">-0.75</Cylinder>
  <Axis unit="deg">30</Axis>
  <ADD unit="D">+1.50</ADD>
  <Prism unit="pri"></Prism>
  <PrismBase unit="deg"></PrismBase>
  <PrismX unit="pri" base="in">1.00</PrismX>
  <PrismY unit="pri" base="down">1.50</PrismY>
  <VA> 0.04-1</VA>
  <ADDVA> 0.04 -1</ADDVA>
  <PHVA>&lt;0.04 </PHVA>
  <LIVA> 0.04-1</LIVA>
  <PD unit="mm">33.0</PD>
</L>
<B>
  <VA>&lt;0.04 </VA>
  <ADDVA>&lt;0.04 </ADDVA>
  <PHVA> 0.04+1</PHVA>
  <LIVA> 0.04+1</LIVA>
  <PD unit="mm">65.5</PD>
  <Div>
    <Blur unit="pri">-21.60</Blur>
    <Break unit="pri">-26.20</Break>
    <Recovery unit="pri">-40.00</Recovery>
  </Div>
  <Conv>
    <Blur unit="pri">-23.80</Blur>
    <Break unit="pri">23.80</Break>
    <Recovery unit="pri">-40.00</Recovery>
  </Conv>
  <Supra>
    <RL>R</RL>
    <Break unit="pri">0.50</Break>
    <Recovery unit="pri">0.20</Recovery>
  </Supra>
  <Infra>
    <RL>R</RL>
    <Break unit="pri">0.40</Break>
    <Recovery unit="pri">0.10</Recovery>
  </Infra>
</B>
</Corrected>
<Corrected CorrectionType="Best" Vision="Near" Situation="Standard">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
<R>
  <Sphere unit="D">-6.75</Sphere>
  <Cylinder unit="D">-1.50</Cylinder>
  <Axis unit="deg">170</Axis>
  <ADD unit="D">-1.50</ADD>
  <Prism unit="pri"></Prism>

```

```

<PrismBase unit="deg"></PrismBase>
<PrismX unit="pri" base="in">1.50</PrismX>
<PrismY unit="pri" base="up">1.50</PrismY>
<VA> 0.04-1</VA>
<ADDVA></ADDVA>
<PHVA> 0.04 -1</PHVA>
<PD unit="mm">30.5</PD>
</R>
<L>
  <Sphere unit="D">-6.25</Sphere>
  <Cylinder unit="D">-0.75</Cylinder>
  <Axis unit="deg">30</Axis>
  <ADD unit="D">-1.50</ADD>
  <Prism unit="pri"></Prism>
  <PrismBase unit="deg"></PrismBase>
  <PrismX unit="pri" base="in">1.50</PrismX>
  <PrismY unit="pri" base="down">1.50</PrismY>
  <VA> 0.05 +1</VA>
  <ADDVA></ADDVA>
  <PHVA> 0.04+1</PHVA>
  <PD unit="mm">31.0</PD>
</L>
<B>
  <VA>&lt;0.04 </VA>
  <ADDVA></ADDVA>
  <PHVA>&lt;0.04 </PHVA>
  <PD unit="mm">61.5</PD>
  <Div>
    <Blur unit="pri">1.20</Blur>
    <Break unit="pri">2.00</Break>
    <Recovery unit="pri">+1.60</Recovery>
  </Div>
  <Conv>
    <Blur unit="pri">1.40</Blur>
    <Break unit="pri">2.60</Break>
    <Recovery unit="pri">+2.20</Recovery>
  </Conv>
  <Supra>
    <RL>R</RL>
    <Break unit="pri">0.10</Break>
    <Recovery unit="pri">-0.40</Recovery>
  </Supra>
  <Infra>
    <RL>R</RL>
    <Break unit="pri">0.40</Break>
    <Recovery unit="pri">0.70</Recovery>
  </Infra>
</B>
</Corrected>
<Corrected CorrectionType="Best" Vision="Distant" Situation="Night">
  <DisplayName></DisplayName>
  <VD unit="mm"></VD>
  <WorkingDistance unit="cm">40</WorkingDistance>
  <R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <ADD unit="D">+1.25</ADD>
    <Prism unit="pri"></Prism>

```

```

<PrismBase unit="deg"></PrismBase>
<PrismX unit="pri" base="in">0.50</PrismX>
<PrismY unit="pri" base="up">0.50</PrismY>
<VA> 0.04 +1</VA>
<ADDVA> 0.04+1</ADDVA>
<PHVA> 2.0 +1</PHVA>
<PD unit="mm">32.5</PD>
</R>
<L>
  <Sphere unit="D">-7.75</Sphere>
  <Cylinder unit="D">-0.75</Cylinder>
  <Axis unit="deg">30</Axis>
  <ADD unit="D">+1.25</ADD>
  <Prism unit="pri"></Prism>
  <PrismBase unit="deg"></PrismBase>
  <PrismX unit="pri" base="in">0.50</PrismX>
  <PrismY unit="pri" base="down">0.50</PrismY>
  <VA> 0.04-1</VA>
  <ADDVA> 0.04 -1</ADDVA>
  <PHVA> 2.0 -1</PHVA>
  <PD unit="mm">33.0</PD>
</L>
<B>
  <VA>&lt;0.04 </VA>
  <ADDVA>&lt;0.04 </ADDVA>
  <PHVA>&lt;0.04 </PHVA>
  <PD unit="mm">65.5</PD>
</B>
</Corrected>
<Other>
  <Fusion>4</Fusion>
  <Stereo>1000X</Stereo>
  <LIStereo>1000 </LIStereo>
  <Aniseikonia>
    <Horizontal>00</Horizontal>
    <Vertical>00</Vertical>
  </Aniseikonia>
  <Dominant_Eye>R</Dominant_Eye>
  <Glare_Contrast>
    <Contrast>100</Contrast>
    <Glare>N</Glare>
  </R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
    <VA> 0.04 +1</VA>
  </R>
  <L>
    <Sphere unit="D">-7.75</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
    <VA> 0.04-1</VA>
  </L>
  <B>
    <VA>&lt;0.04 </VA>
  </B>
</Glare_Contrast>
<ETDRS>
  <Contrast>6</Contrast>

```

```

<Glare>N</Glare>
<R>
  <Sphere unit="D">-8.25</Sphere>
  <Cylinder unit="D">-1.50</Cylinder>
  <Axis unit="deg">170</Axis>
  <VA> 12.5 +2</VA>
</R>
<L>
  <Sphere unit="D">-7.75</Sphere>
  <Cylinder unit="D">-0.75</Cylinder>
  <Axis unit="deg">30</Axis>
  <VA> 10 -2</VA>
</L>
<B>
  <VA> 12.5 +2</VA>
</B>
</ETDRS>
</Other>
</Phoropter>
<The21pointExams>
<No07>
  <R>
    <Sphere unit="D">-8.25</Sphere>
    <Cylinder unit="D">-1.50</Cylinder>
    <Axis unit="deg">170</Axis>
  </R>
  <L>
    <Sphere unit="D">-7.75</Sphere>
    <Cylinder unit="D">-0.75</Cylinder>
    <Axis unit="deg">30</Axis>
  </L>
</No07>
<No0910>
  <Blur unit="pri">0.40</Blur>
  <Break unit="pri">0.80</Break>
  <Recovery unit="pri">+1.20</Recovery>
</No0910>
<No11>
  <Blur unit="pri">-0.60</Blur>
  <Break unit="pri">-1.20</Break>
  <Recovery unit="pri">-1.80</Recovery>
</No11>
<No12B>
  <Supra>
    <RL>R</RL>
    <Break unit="pri">0.60</Break>
    <Recovery unit="pri">-0.20</Recovery>
  </Supra>
  <Infra>
    <RL>R</RL>
    <Break unit="pri">0.40</Break>
    <Recovery unit="pri">-0.60</Recovery>
  </Infra>
</No12B>
<No14A>
  <R>
    <ADD unit="D">-0.50</ADD>
  </R>
  <L>

```

```

    <ADD unit="D">-0.75</ADD>
  </L>
</No14A>
<No14B>
  <R>
    <ADD unit="D">-0.50</ADD>
  </R>
  <L>
    <ADD unit="D">-0.50</ADD>
  </L>
</No14B>
<No15A>
  <PrismH unit="pri" phoria="Exo">2.00</PrismH>
</No15A>
<No15B>
  <PrismH unit="pri" phoria="Exo">3.00</PrismH>
</No15B>
<No16>
  <Blur unit="pri">-0.40</Blur>
  <Break unit="pri">-0.80</Break>
  <Recovery unit="pri">-1.20</Recovery>
</No16>
<No17>
  <Blur unit="pri">0.40</Blur>
  <Break unit="pri">0.80</Break>
  <Recovery unit="pri">+1.20</Recovery>
</No17>
<No18B>
  <Supra>
    <RL>R</RL>
    <Break unit="pri">0.40</Break>
    <Recovery unit="pri">0.90</Recovery>
  </Supra>
  <Infra>
    <RL>R</RL>
    <Break unit="pri">0.80</Break>
    <Recovery unit="pri">1.40</Recovery>
  </Infra>
</No18B>
<No19>
  <MinusLens>
    <R>
      <ACC unit="D">+2.75</ACC>
    </R>
    <L>
      <ACC unit="D">+2.50</ACC>
    </L>
    <B>
      <ACC unit="D">+2.25</ACC>
    </B>
  </MinusLens>
  <NPA>
    <R>
      <ACC unit="D">3.26</ACC>
      <Distance unit="cm">21</Distance>
    </R>
    <L>
      <ACC unit="D">3.50</ACC>
      <Distance unit="cm">20</Distance>
    </L>
  </NPA>

```

```

</L>
<B>
  <ACC unit="D">3.76</ACC>
  <Distance unit="cm">19</Distance>
</B>
</NPA>
</No19>
<No20>
  <R>
    <Blur unit="D">-0.75</Blur>
    <Recovery unit="D">-0.25</Recovery>
  </R>
  <L>
    <Blur unit="D">-1.25</Blur>
    <Recovery unit="D">-0.75</Recovery>
  </L>
  <B>
    <Blur unit="D">-0.75</Blur>
    <Recovery unit="D">-0.25</Recovery>
  </B>
</No20>
<No21>
  <R>
    <Blur unit="D">+1.00</Blur>
    <Recovery unit="D">+0.50</Recovery>
  </R>
  <L>
    <Blur unit="D">+0.75</Blur>
    <Recovery unit="D">+0.25</Recovery>
  </L>
  <B>
    <Blur unit="D">+0.50</Blur>
    <Recovery unit="D">+0.25</Recovery>
  </B>
</No21>
<NPC>
  <Distance unit="cm">45</Distance>
  <MA unit="MA">2.1</MA>
  <Prism unit="pri">13.8</Prism>
</NPC>
<GradACA>
  <ACA> 0.00</ACA>
  <Sheard>
    <Distant>
      <PrismX unit="pri" base="out">0.30</PrismX>
      <Sphere unit="D"></Sphere>
    </Distant>
    <Near>
      <PrismX unit="pri" base="in">0.60</PrismX>
      <Sphere unit="D"></Sphere>
    </Near>
  </Sheard>
  <Percival>
    <Distant>
      <PrismX unit="pri" base="in">0.20</PrismX>
      <Sphere unit="D"></Sphere>
    </Distant>
    <Near>
      <PrismX unit="pri" base="">0.00</PrismX>

```

```

    <Sphere unit="D"></Sphere>
  </Near>
</Percival>
</GradACA>
<HeteroACA>
  <ACA>14.37</ACA>
</HeteroACA>
<Morgan>
  <A>LLLLL</A>
  <B>LLLLLLL</B>
  <C>LNL</C>
</Morgan>
</The21pointExams>
<KM>
  <ReferencedInstance></ReferencedInstance>
  <DiopterStep unit="D"></DiopterStep>
  <AxisStep unit="deg"></AxisStep>
  <CylinderMode></CylinderMode>
  <R>
    <Median>
      <R1>
        <Radius unit="mm">7.75</Radius>
        <Power unit="D">43.50</Power>
        <Axis unit="deg">165</Axis>
      </R1>
      <R2>
        <Radius unit="mm">7.51</Radius>
        <Power unit="D">45.00</Power>
        <Axis unit="deg"></Axis>
      </R2>
    </Median>
  </R>
  <L>
    <Median>
      <R1>
        <Radius unit="mm">7.79</Radius>
        <Power unit="D">43.25</Power>
        <Axis unit="deg">20</Axis>
      </R1>
      <R2>
        <Radius unit="mm">7.64</Radius>
        <Power unit="D">44.25</Power>
        <Axis unit="deg"></Axis>
      </R2>
    </Median>
  </L>
</KM>
<TM>
  <ReferencedInstance></ReferencedInstance>
  <R>
    <Average>
      <IOP_mmHg unit="mmHg">16.8</IOP_mmHg>
      <IOP_Pa unit="kPa">2.24</IOP_Pa>
    </Average>
  </R>
  <L>
    <Average>
      <IOP_mmHg unit="mmHg">17.4</IOP_mmHg>
      <IOP_Pa unit="kPa">2.33</IOP_Pa>

```

```
</Average>
</L>
</TM>
<Test_Time unit="sec">1196</Test_Time>
<SystemNo>3</SystemNo>
</Measure>
</Ophthalmology>
```

3.1.4 Serial data

For the serial data output to the computer via the MEM-200, refer to the RT-5100 Interface Manual.

3.2 Data Input from Computer

For the serial data input from the computer via the MEM-200, refer to the RT-5100 Interface Manual.

